

Scaling Instruction-Finetuned Language Models

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Google Research

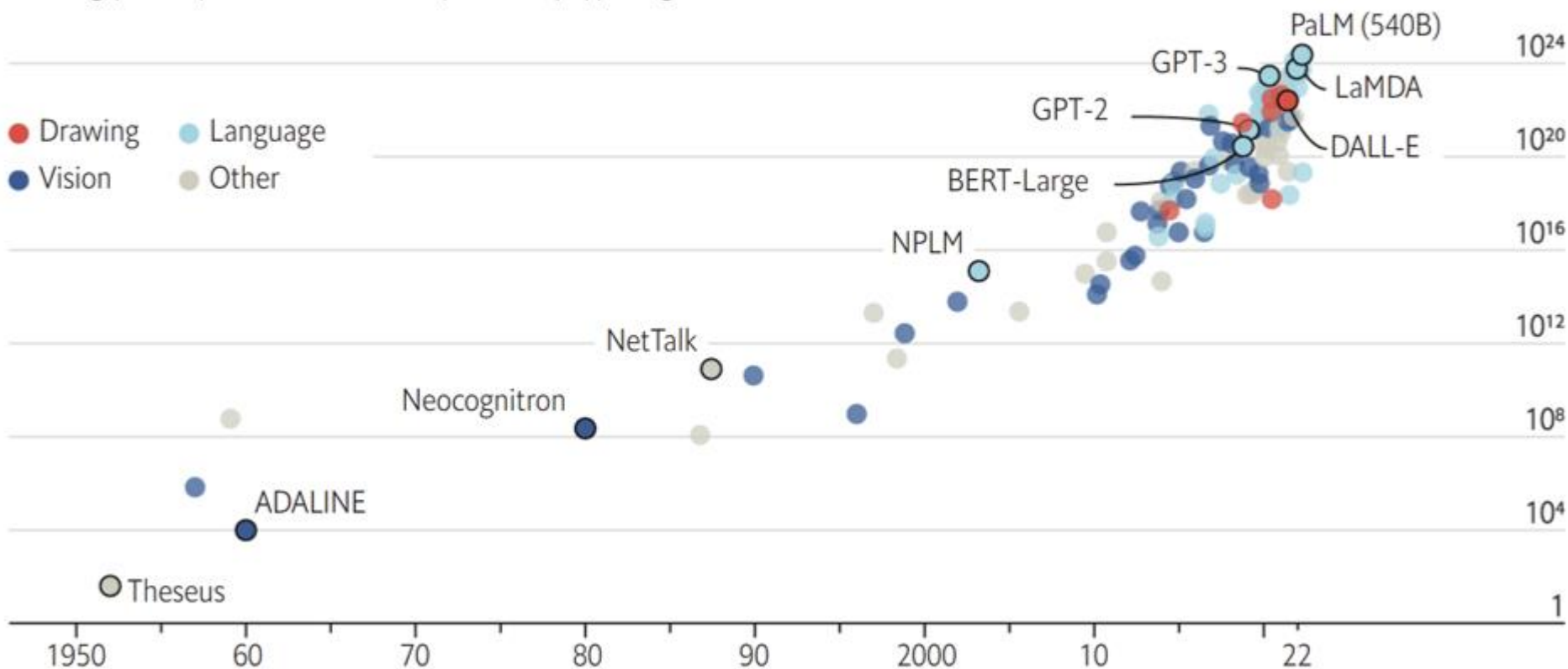
arXiv:2210.11416 (2022)

Presenter : 박산희, 조해창, 변현정, 이세현

Motivation

The blessings of scale

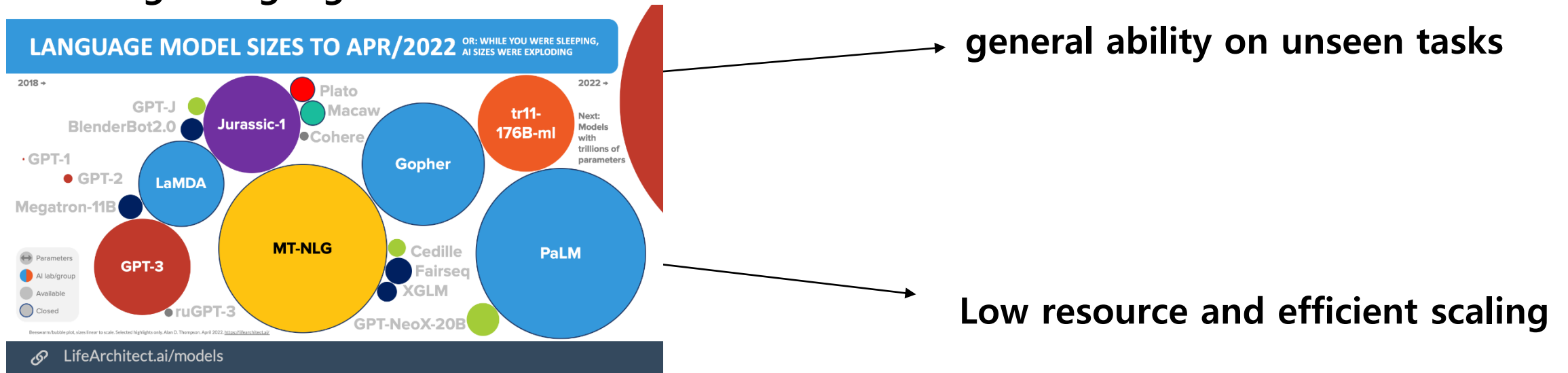
AI training runs, estimated computing resources used
Floating-point operations, selected systems, by type, log scale



Sources: "Compute trends across three eras of machine learning", by J. Sevilla et al., arXiv, 2022; Our World in Data

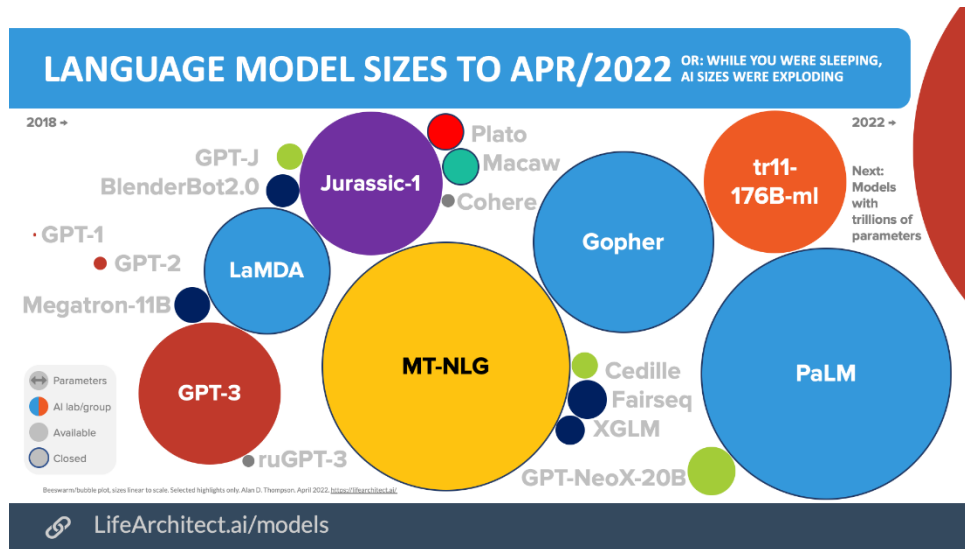
Motivation

Large Language Models

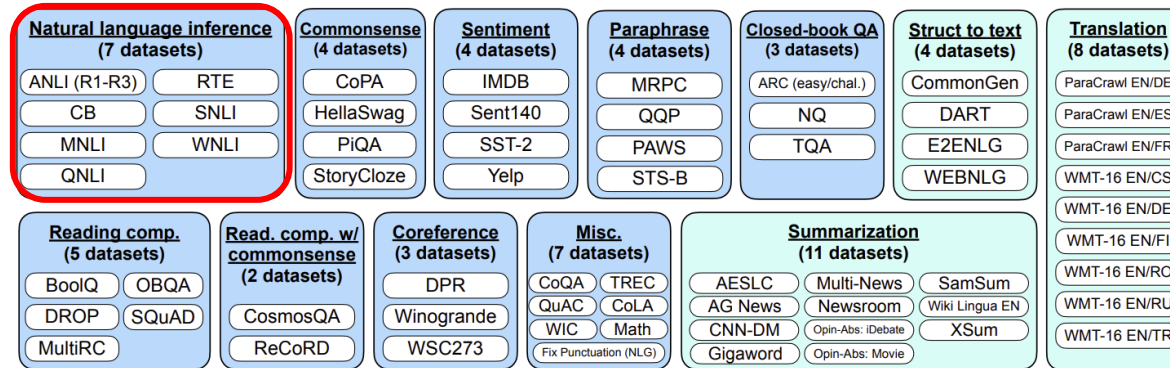


Motivation

Large Language Models



A mixture



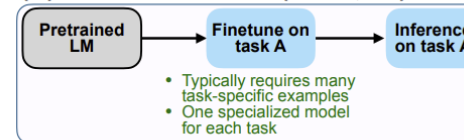
Held-out Tasks

general ability on unseen tasks

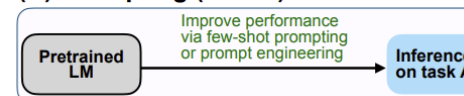
Low resource and efficient scaling

Pathway ..., x-shot Tuning

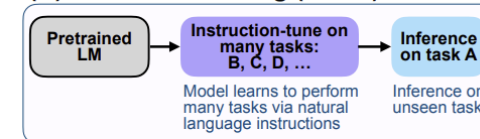
(A) Pretrain-finetune (BERT, T5)



(B) Prompting (GPT-3)

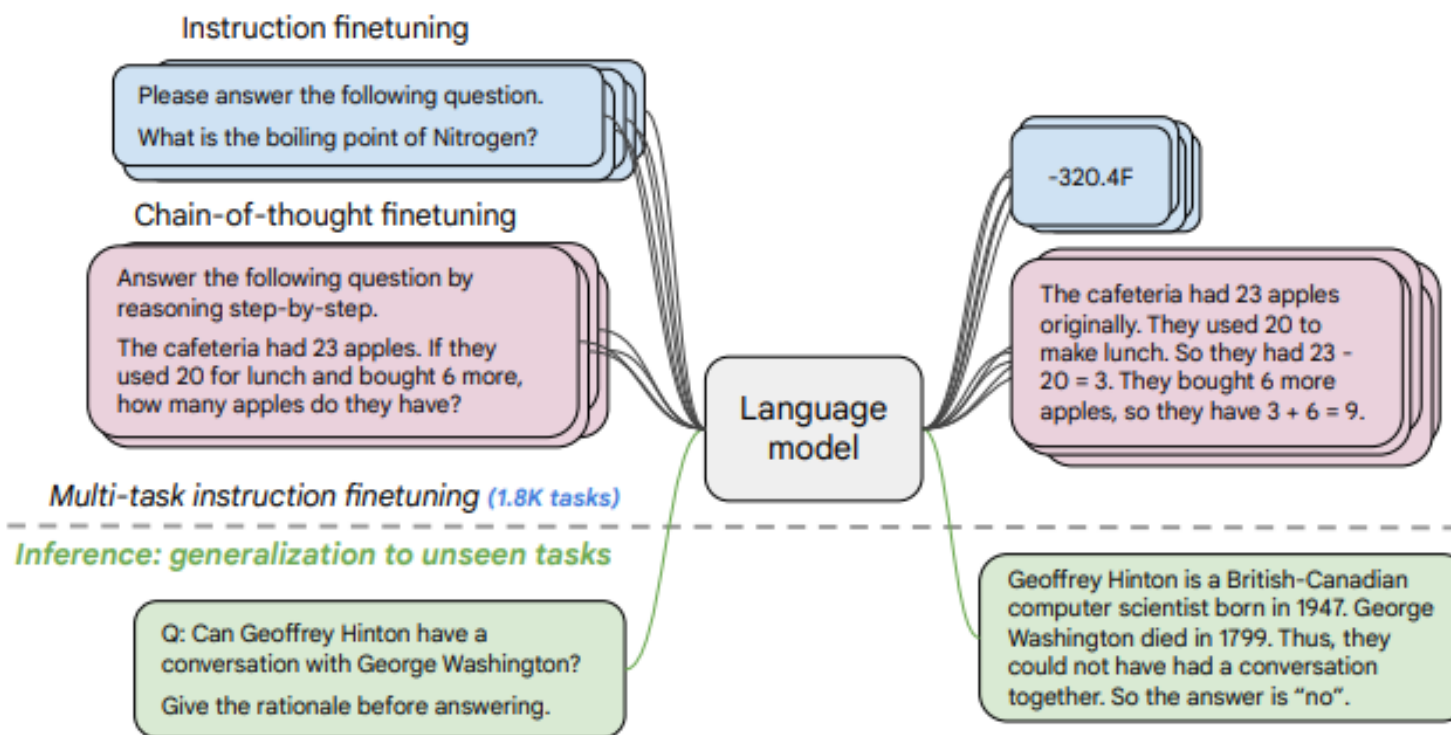


(C) Instruction tuning (FLAN)



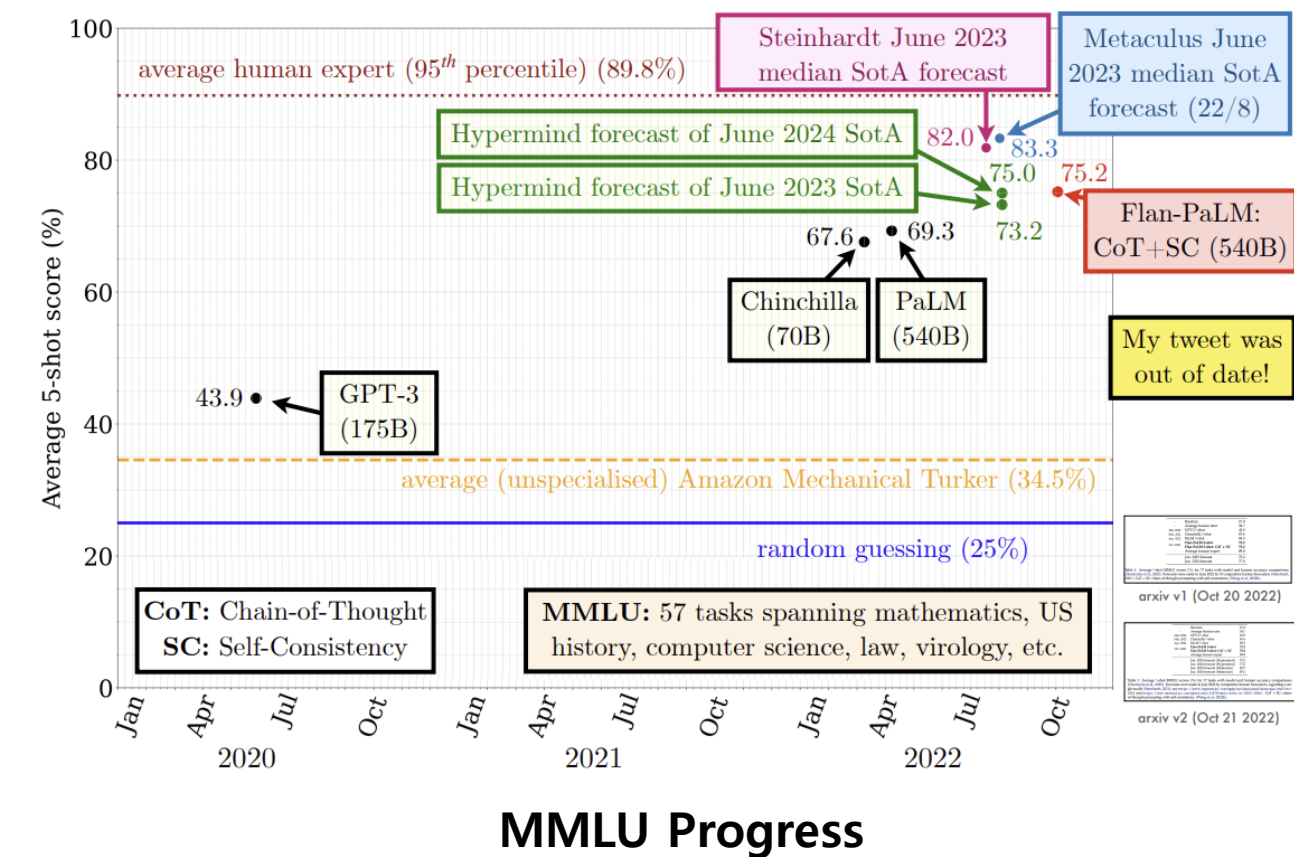
Introduction

In AI, generalized model to **answer the unseen tasks** is an important goal. Finetuning language models on a collection of tasks phrased as instructions enables model to respond better to instructions and reduces the need for few-shot exemplars.



Introduction

The need for scaling on instruction finetuning : number of tasks and the size of the model

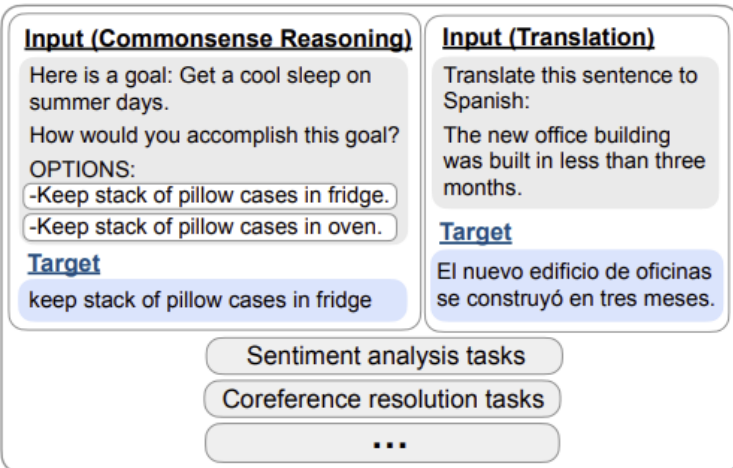


-	Random	25.0
-	Average human rater	34.5
May 2020	GPT-3 5-shot	43.9
Mar. 2022	Chinchilla 5-shot	67.6
Apr. 2022	PaLM 5-shot	69.3
Oct. 2022	Flan-PaLM 5-shot	72.2
	Flan-PaLM 5-shot: CoT + SC	75.2
-	Average human expert	89.8
	Jun. 2023 forecast (Hypermind)	73.2
	Jun. 2024 forecast (Hypermind)	75.0
	Jun. 2023 forecast (Metaculus)	82.7
	Jun. 2024 forecast (Metaculus)	87.6

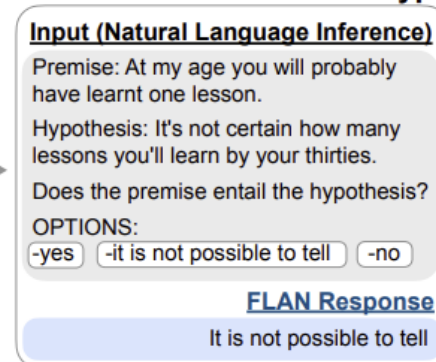
FLAN

(Jason Wei et al., ICLR 2022)

Finetune on many tasks (“instruction-tuning”)

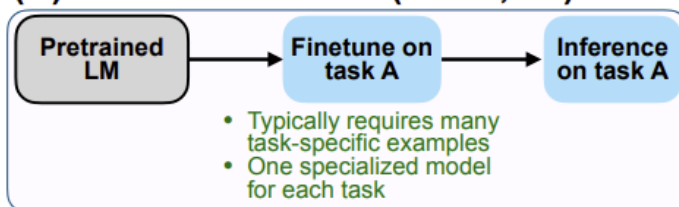


Inference on unseen task type

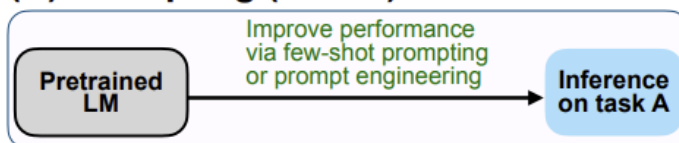


Instruction tuning: finetuning language models on a collection of datasets described via instructions

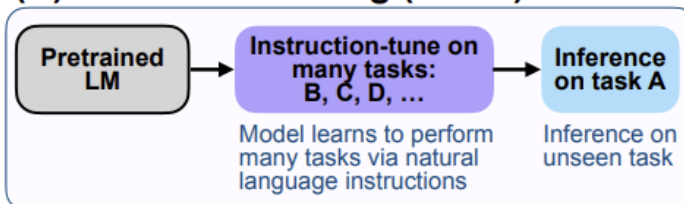
(A) Pretrain–finetune (BERT, T5)



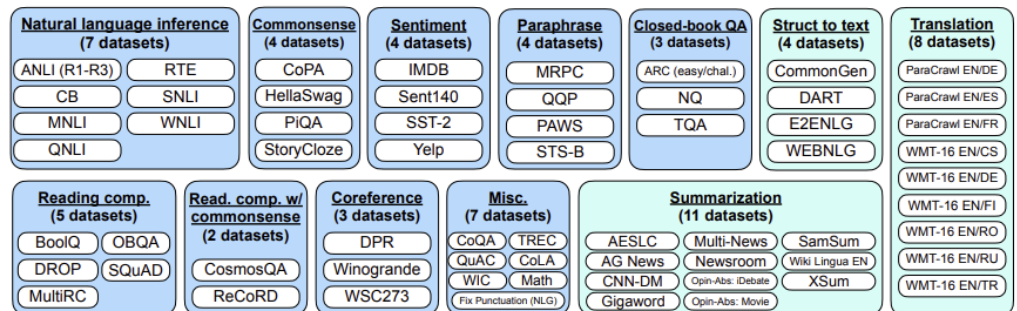
(B) Prompting (GPT-3)



(C) Instruction tuning (FLAN)

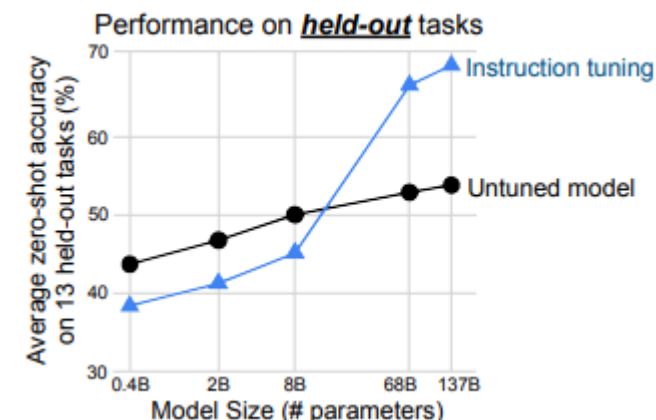
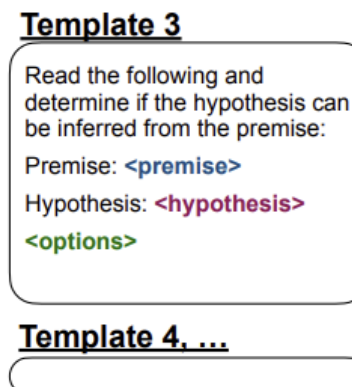
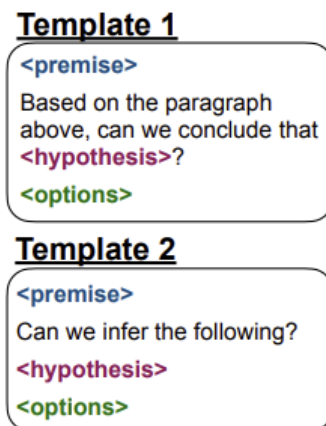
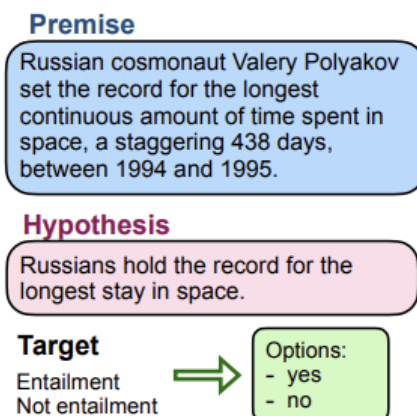
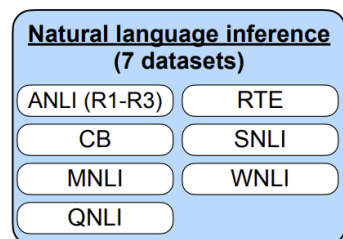


FLAN (Jason Wei et al., ICLR 2022)



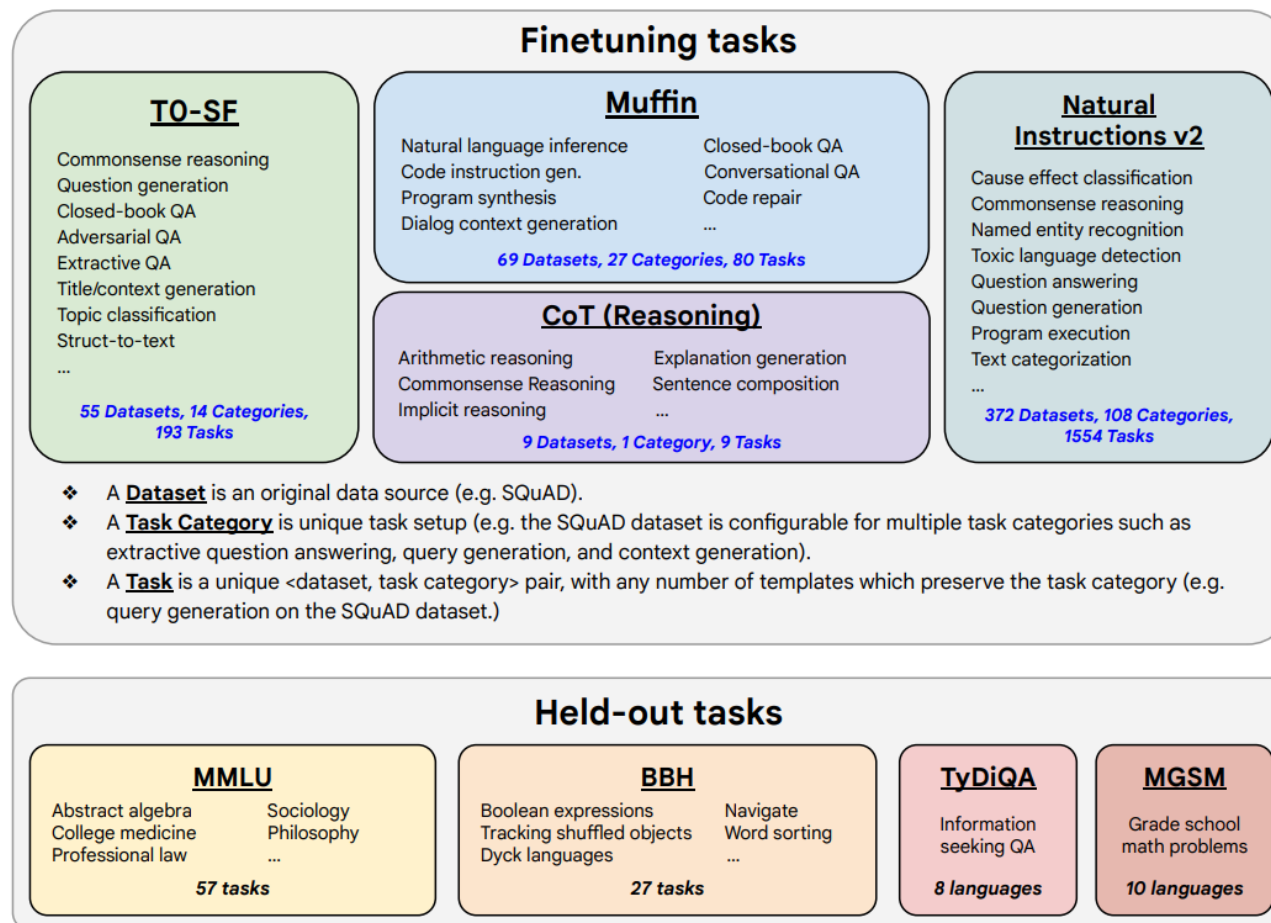
A pretrained language model is Instruction tune on the **mixture of all datasets** with examples in each dataset.

Mixture



Flan Finetuning

473 Datasets, 146 task categories, 1,836 total tasks : combination tasks from Flan (ICLR 2022), T0, Natural Instruction.



Flan Finetuning

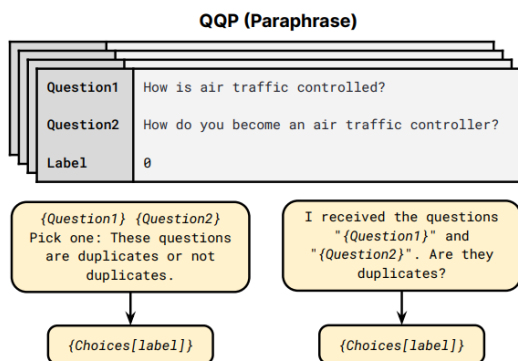
473 Datasets, 146 task categories, 1,836 total tasks : combination tasks from Flan (ICLR 2022), T0, Natural Instruction

FLAN prompt:

Premise: At my age you will probably have learnt one lesson.

Hypothesis: It's not certain how many lessons you'll learn by your thirties.

Does the premise entail the hypothesis?



Finetuning tasks

T0-SF

Commonsense reasoning
Question generation
Closed-book QA
Adversarial QA
Extractive QA
Title/context generation
Topic classification
Struct-to-text
...

55 Datasets, 14 Categories,
193 Tasks

Muffin

Natural language inference
Code instruction gen.
Program synthesis
Dialog context generation
Closed-book QA
Conversational QA
Code repair
...

69 Datasets, 27 Categories, 80 Tasks

CoT (Reasoning)

Arithmetic reasoning
Commonsense Reasoning
Implicit reasoning
Explanation generation
Sentence composition
...

9 Datasets, 1 Category, 9 Tasks

Natural Instructions v2

Cause effect classification
Commonsense reasoning
Named entity recognition
Toxic language detection
Question answering
Question generation
Program execution
Text categorization
...

372 Datasets, 108 Categories,
1554 Tasks

- ❖ A **Dataset** is an original data source (e.g. SQuAD).
- ❖ A **Task Category** is unique task setup (e.g. the SQuAD dataset is configurable for multiple task categories such as extractive question answering, query generation, and context generation).
- ❖ A **Task** is a unique <dataset, task category> pair, with any number of templates which preserve the task category (e.g. query generation on the SQuAD dataset.)

Positive Examples

- **Input:** "Context: ... 'That's fantastic, I'm glad we came to something we both agree with.' Utterance: 'Me too. I hope you have a wonderful camping trip.'"
- **Output:** "Yes"
- **Explanation:** "The participant engages in small talk when wishing their opponent to have a wonderful trip."

Held-out tasks

MMLU

Abstract algebra
College medicine
Professional law
Sociology
Philosophy
...

57 tasks

BBH

Boolean expressions
Tracking shuffled objects
Dyck languages
Navigate
Word sorting
...

27 tasks

TyDiQA

Information seeking QA

8 languages

MGSM

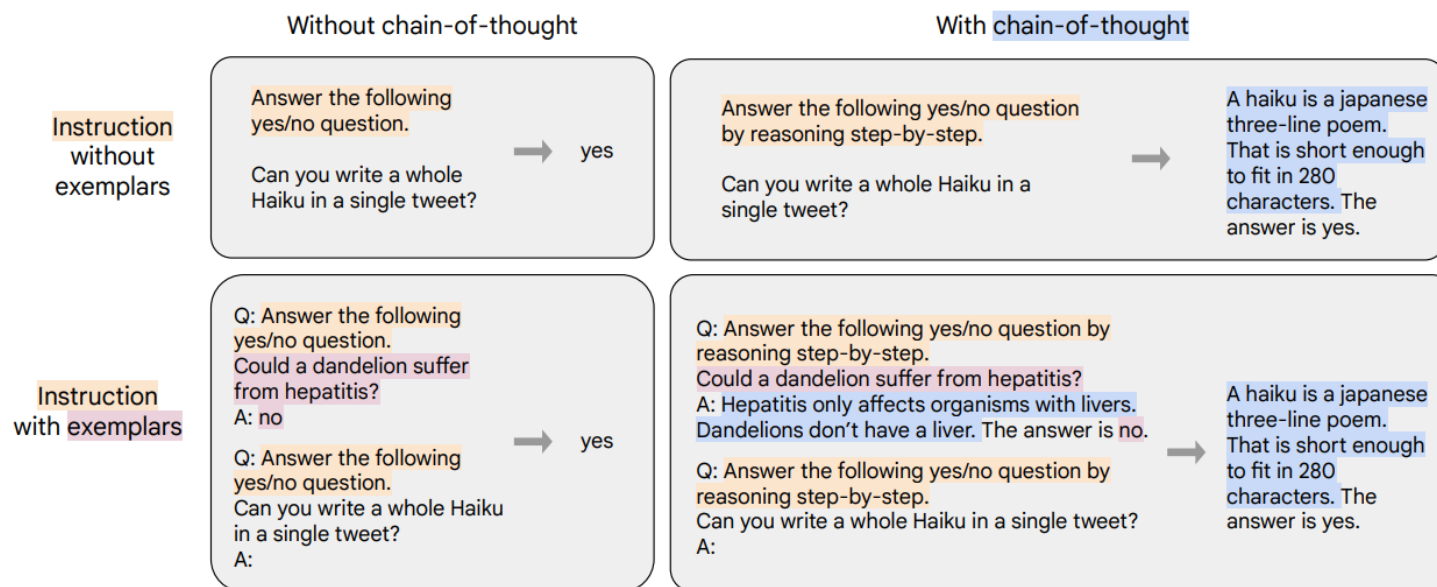
Grade school math problems

10 languages

Flan Finetuning

Chain-of-thought finetuning mixture: dataset formats compose of four combinations : w w/o exemplars & w w/o CoT

→ Finetuning on CoT annotation improves performance on unseen reasoning tasks.



Flan Finetuning

Fine tuning procedure : T5, PaLM, U-PaLM

Params	Model	Architecture	Pre-training Objective	Pre-train FLOPs	Finetune FLOPs	% Finetune Compute
80M	Flan-T5-Small	encoder-decoder	span corruption	1.8E+20	2.9E+18	1.6%
250M	Flan-T5-Base	encoder-decoder	span corruption	6.6E+20	9.1E+18	1.4%
780M	Flan-T5-Large	encoder-decoder	span corruption	2.3E+21	2.4E+19	1.1%
3B	Flan-T5-XL	encoder-decoder	span corruption	9.0E+21	5.6E+19	0.6%
11B	Flan-T5-XXL	encoder-decoder	span corruption	3.3E+22	7.6E+19	0.2%
8B	Flan-PaLM	decoder-only	causal LM	3.7E+22	1.6E+20	0.4%
62B	Flan-PaLM	decoder-only	causal LM	2.9E+23	1.2E+21	0.4%
540B	Flan-PaLM	decoder-only	causal LM	2.5E+24	5.6E+21	0.2%
62B	Flan-cont-PaLM	decoder-only	causal LM	4.8E+23	1.8E+21	0.4%
540B	Flan-U-PaLM	decoder-only	prefix LM + span corruption	2.5E+23	5.6E+21	0.2%

Params	Model	Batch size	Dropout	LR	Steps
80M	Flan-T5-Small	64	0.05	5e-4	98k
250M	Flan-T5-Base	64	0.05	5e-4	84k
780M	Flan-T5-Large	64	0.05	5e-4	64k
3B	Flan-T5-XL	64	0.05	5e-4	38k
11B	Flan-T5-XXL	64	0.05	5e-4	14k
8B	Flan-PaLM	32	0.05	3e-3	40k
62B	Flan-PaLM	32	0.05	3e-3	40k
540B	Flan-PaLM	32	0.1	1e-3	21k
62B	Flan-cont-PaLM	32	0.05	3e-3	60k
540B	Flan-U-PaLM	32	0.1	1e-3	30k

Table 22: Hyperparameters used for all finetuned models.

Mixture	Maximum cap	Proportion (A)	Proportion (B)
Muffin	30,000	52%	46.0%
T0-SF	20,000	15%	27.9%
CoT	100,000	3%	1.8%
NIV2	5,000	30%	24.2%

Evaluation Protocol

MMLU, BBH, TyDiQA, MGSM

MMLU (Multi Massive Multitask Language Understanding)

Various branch of **knowledge (humanities, social sciences, hard sciences, and other areas)** that are important for some people to learn) 57 tasks, **15908 questions**.

Human-level accuracy is 34.5% (unspecialized humans : AMT), 89.8% (expert-level human)

Microeconomics	One of the reasons that the government discourages and regulates monopolies is that	
	(A) producer surplus is lost and consumer surplus is gained.	✗
	(B) monopoly prices ensure productive efficiency but cost society allocative efficiency.	✗
	(C) monopoly firms do not engage in significant research and development.	✗
	(D) consumer surplus is lost with higher prices and lower levels of output.	✓

Figure 3: Examples from the Microeconomics task.

BIG Benchmark (Beyond the Imitation Game **Benchmark**)

In traditional NLP, 200 diverse text-based tasks such as mathematics, commonsense reasoning, and question-answering. BIG-Bench Hard (BBH) is a subset of 23 particularly **challenging BIG-Bench tasks**.

Multiple-choice qa, open-domain qa, multi-label classification etc.

Evaluation Protocol

MMLU, BBH, TyDiQA, MGSM

TyDiQA

Question answering across 8 typologically diverse languages

Q: من هو موزارت ؟
mn hw mwzArt ?
Who is Muzart?
A: أماديوس موتسارت
>mAdyws mwtsArt
...Amadeus Mozart ...

MGSM

Multilingual benchmark of mathematics problems translated to 10 languages

Question: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

Answer: 11

Frage: Es waren neun Computer im Serverraum. Von Montag bis Donnerstag wurden jeden Tag noch fünf Computer installiert. Wie viele Computer sind jetzt im Serverraum?

Antwort: 29

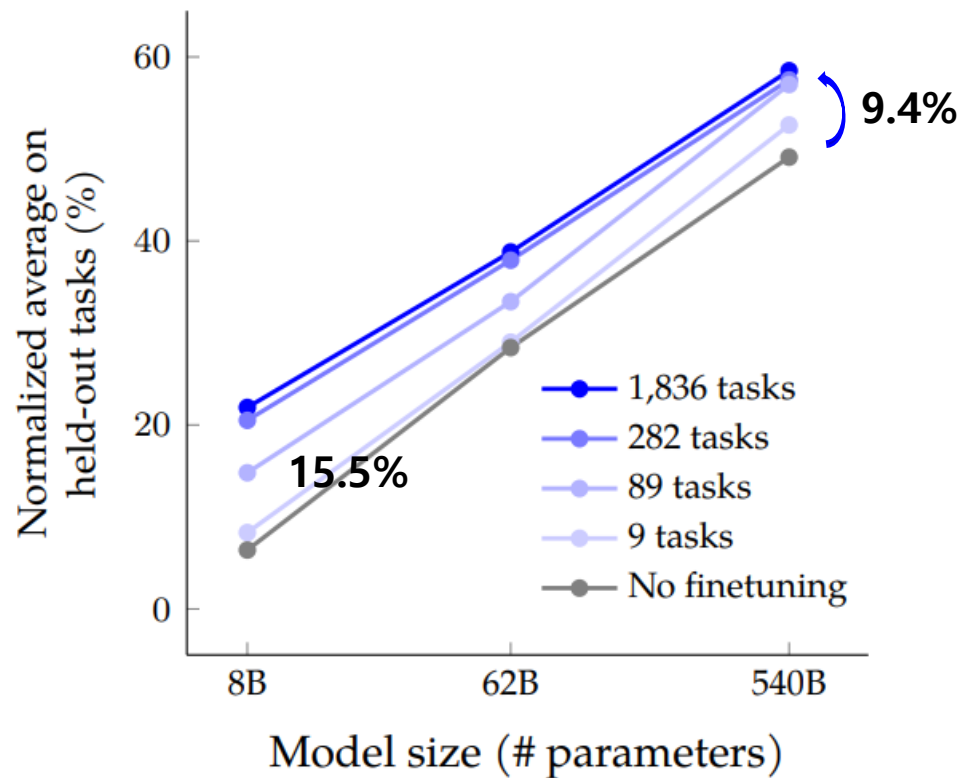
Question: Léa avait 32 chocolats et sa sœur en avait 42. Si elles en ont mangé 35, combien de morceaux leur reste-t-il en tout ?

Réponse: 39

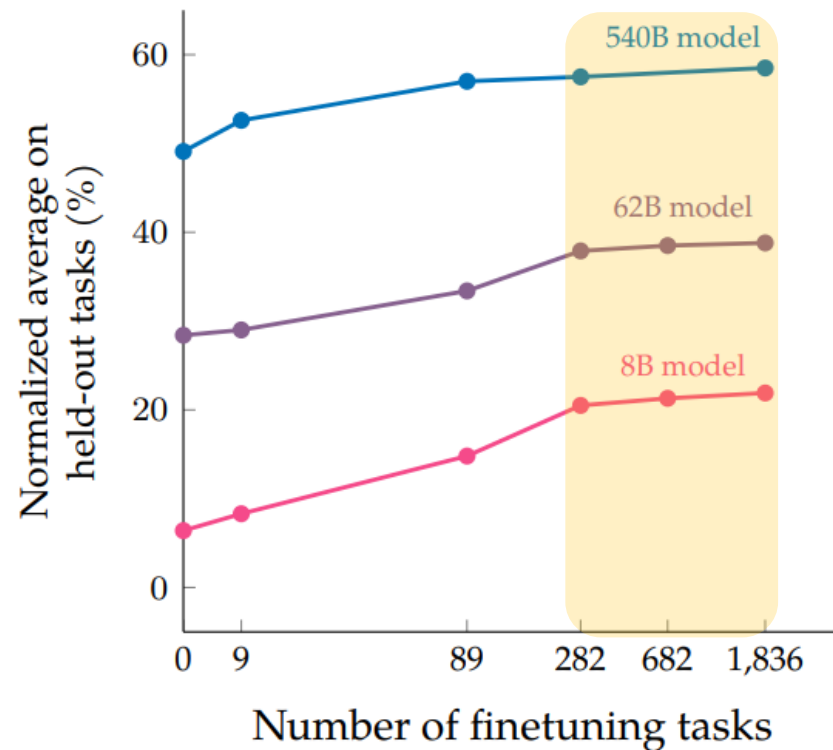
Scaling to 540B parameters and 1.8K tasks

Experiments on three PaLM model sizes

Evaluation metric is few-shot prompted accuracy (exact match)



Scaling the model size even further might improve performance,



Scaling number of finetuning tasks also improve performance, but small gain after 282 tasks.

- Conjectures : Additional tasks are not diverse,
- Model already gains new knowledge from pretraining.

Scaling to 540B parameters and 1.8K tasks

Model	Finetuning Mixtures	Tasks	Norm. avg.	MMLU		BBH		TyDiQA	MGSM
				Direct	CoT	Direct	CoT	Direct	CoT
8B	None (no finetuning)	0	6.4	24.3	24.1	30.8	30.1	25.0	3.4
	CoT	9	8.3 (+1.9)	26.3	32.1	19.8	26.6	39.3	<u>10.4</u>
	CoT, Muffin	89	14.8 (+8.4)	37.6	38.4	31.0	30.9	32.4	8.4
	CoT, Muffin, T0-SF	282	20.5 (+14.1)	47.7	39.7	33.1	30.9	<u>49.0</u>	8.5
	CoT, Muffin, T0-SF, NIV2	1,836	<u>21.9</u> (+15.5)	<u>49.3</u>	<u>41.3</u>	<u>36.4</u>	<u>31.1</u>	47.5	8.2
62B	None (no finetuning)	0	28.4	55.1	49.0	37.4	43.0	40.5	18.2
	CoT	9	29.0 (+0.4)	48.5	48.7	34.5	39.5	48.8	<u>32.6</u>
	CoT, Muffin	89	33.4 (+6.0)	55.3	51.4	42.8	40.2	53.0	23.9
	CoT, Muffin, T0-SF	282	37.9 (+9.5)	<u>60.0</u>	56.0	44.7	43.8	58.2	30.0
	CoT, Muffin, T0-SF, NIV2	1,836	<u>38.8</u> (+10.4)	59.6	<u>56.9</u>	<u>47.5</u>	<u>44.9</u>	<u>58.7</u>	28.5
540B	None (no finetuning)	0	49.1	71.3	62.9	49.1	63.7	52.9	45.9
	CoT	9	52.6 (+3.5)	68.8	64.8	50.5	61.1	61.2	59.4
	CoT, Muffin	89	57.0 (+7.9)	71.8	66.7	56.7	64.0	65.3	<u>63.0</u>
	CoT, Muffin, T0-SF	282	57.5 (+8.4)	72.9	<u>68.2</u>	57.3	64.0	65.8	61.6
	CoT, Muffin, T0-SF, NIV2	1,836	<u>58.5</u> (+9.4)	<u>73.2</u>	68.1	<u>58.8</u>	<u>65.6</u>	<u>67.4</u>	61.3

CoT -> Muffin -> T0-SF-> NIV2 : CoT tuning has most effect on MGSM like world math problem.

4 Finetuning with chain-of-thought annotations

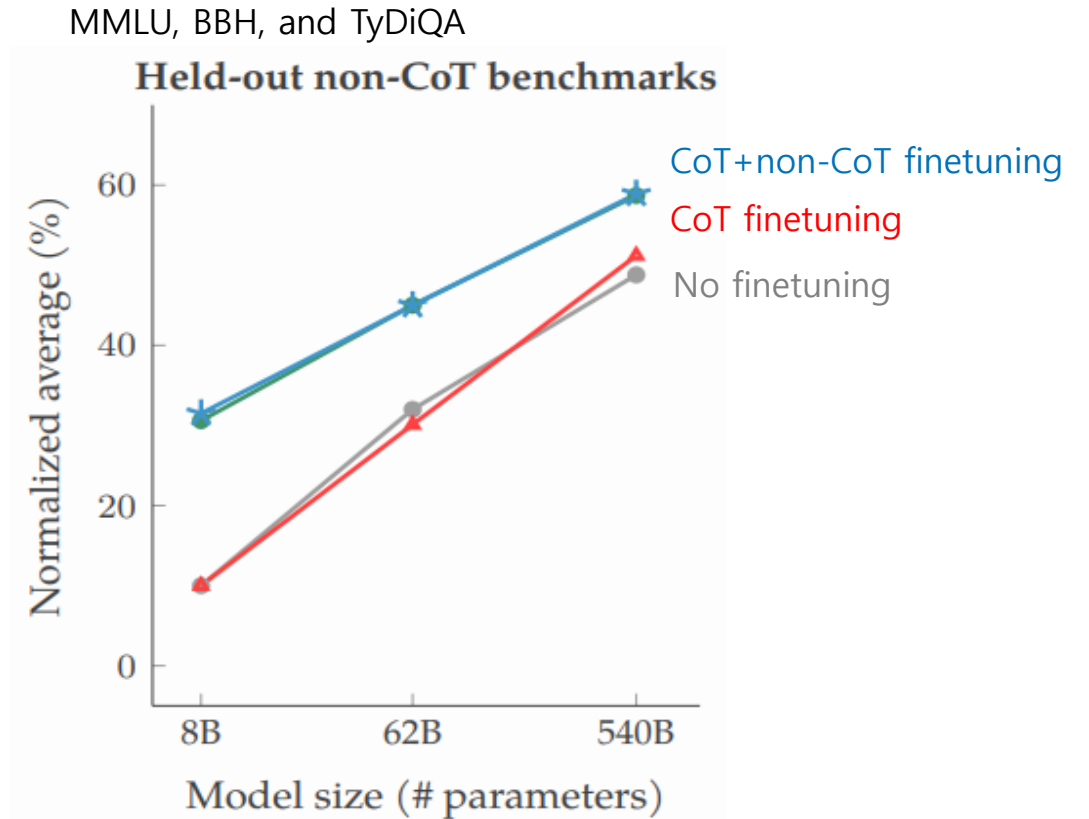
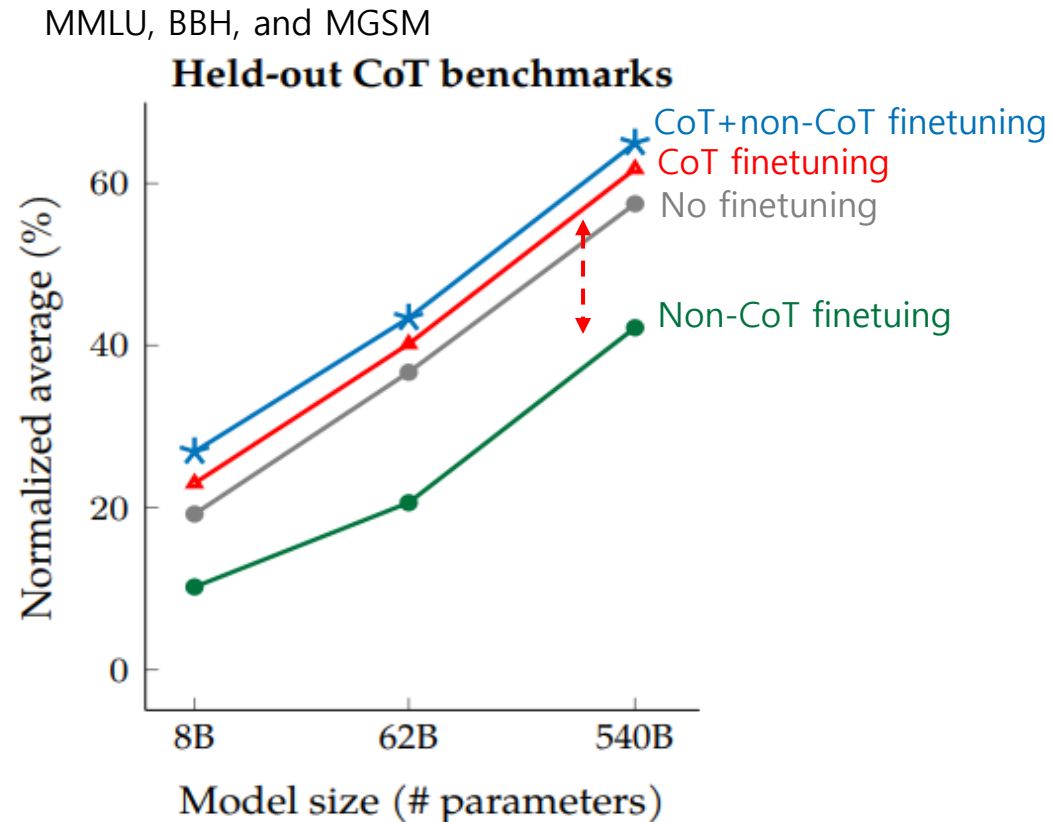
Reasoning ability with chain-of-thought data

	MMLU	BBH-nlp	BBH-alg	TyDiQA	MGSM
Prior best	69.3 ^a	73.5 ^b	<u>73.9</u> ^b	<u>81.9</u> ^c	55.0 ^d
PaLM 540B					
- direct prompting	69.3	62.7	38.3	52.9	18.3
- CoT prompting	64.5	71.2	57.6	-	45.9
- CoT + self-consistency	69.5	78.2	62.2	-	57.9
Flan-PaLM 540B					
- direct prompting	72.2	70.0	48.2	67.8	21.2
- CoT prompting	70.2	72.4	61.3	-	57.0
- CoT + self-consistency	<u>75.2</u>	<u>78.4</u>	66.5	-	<u>72.0</u>

New state-of-the-art

4 Finetuning with chain-of-thought annotations

Reasoning ability with chain-of-thought data



They stratify evaluation into held-out CoT benchmarks and held-out non-CoT benchmarks. It is critical to finetune on some CoT examples to keep reasoning abilities.

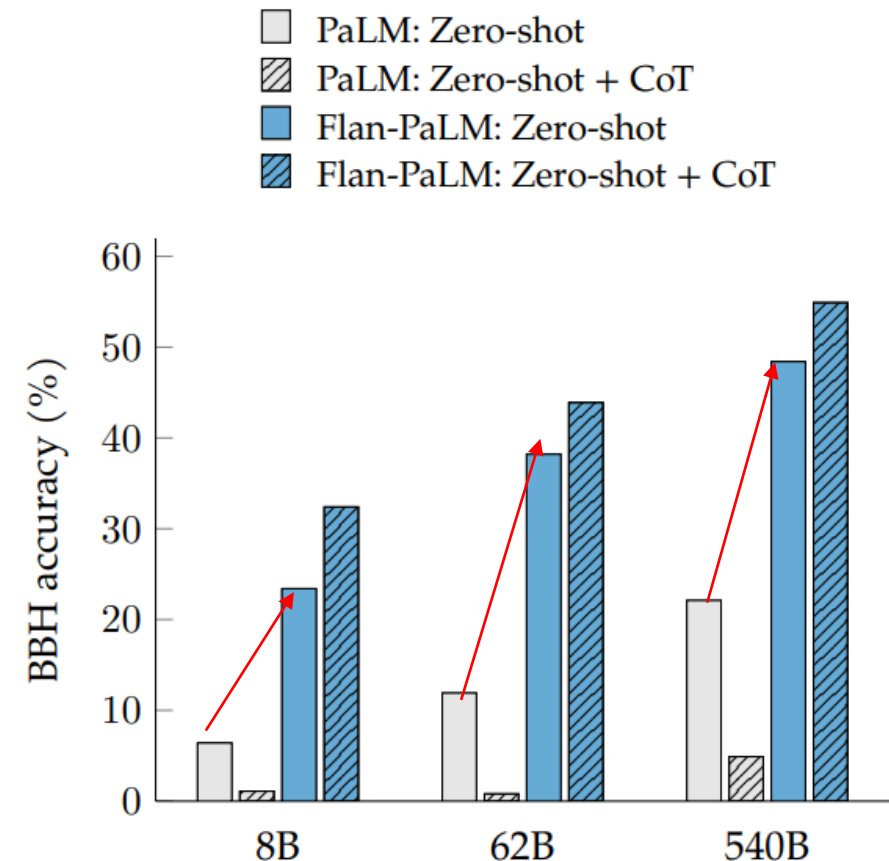
4 Finetuning with chain-of-thought annotations

Unlocking zero-shot reasoning

Instruction finetuning on CoT data both with and without exemplars shows better performance of CoT reasoning in a zero-shot setting.

In BBH, finetuning Flan-PaLM with some CoT dataset enables zero-shot CoT reasoning in unseen tasks.

Model input (Boolean Expressions) Q: (False or not False or False) is A: Let's think step by step.	PaLM 540B output (False or not False or False) (False or False or False) (False or False) (False) ❌ (wrong answer)	Flan-PaLM 540B output False or not False is True. True or False is True. The answer: True. ✅
Model input (Disambiguation QA) Q: In the following sentences, explain the antecedent of the pronoun (which thing the pronoun refers to), or state that it is ambiguous. Sentence: The reporter and the chef will discuss their favorite dishes. Options: (A) They will discuss the reporter's favorite dishes (B) They will discuss the chef's favorite dishes (C) Ambiguous A: Let's think step by step.	PaLM 540B output The reporter and the chef will discuss their favorite dishes. The reporter and the chef will discuss the reporter's favorite dishes. The reporter and the chef will discuss the chef's favorite dishes. The reporter and the chef will discuss the reporter's and the chef's favorite dishes. ❌ (doesn't answer question)	Flan-PaLM 540B output The reporter and the chef will discuss their favorite dishes does not indicate whose favorite dishes they will discuss. So, the answer is (C). ✅



Putting it all together

GPT-3
175B : >43.9% →

Instruction finetuning's
combination best
achievement →

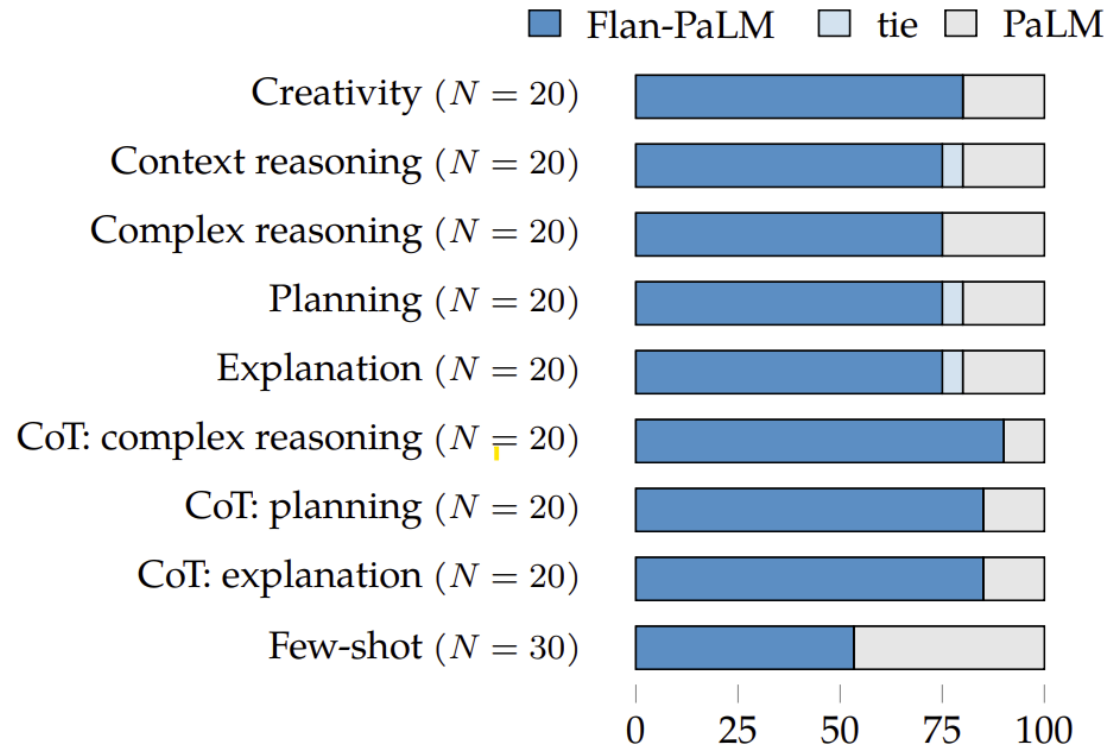
Params	Model	Norm. avg.	MMLU		BBH		TyDiQA	MGSM
			Direct	CoT	Direct	CoT	Direct	CoT
80M	T5-Small	-9.2	26.7	5.6	27.0	7.2	0.0	0.4
	Flan-T5-Small	-3.1 (+6.1)	28.7	12.1	29.1	19.2	1.1	0.2
250M	T5-Base	-5.1	25.7	14.5	27.8	14.6	0.0	0.5
	Flan-T5-Base	6.5 (+11.6)	35.9	33.7	31.3	27.9	4.1	0.4
780M	T5-Large	-5.0	25.1	15.0	27.7	16.1	0.0	0.3
	Flan-T5-Large	13.8 (+18.8)	45.1	40.5	37.5	31.5	12.3	0.7
3B	T5-XL	-4.1	25.7	14.5	27.4	19.2	0.0	0.8
	Flan-T5-XL	19.1 (+23.2)	52.4	45.5	41.0	35.2	16.6	1.9
11B	T5-XXL	-2.9	25.9	18.7	29.5	19.3	0.0	1.0
	Flan-T5-XXL	23.7 (+26.6)	55.1	48.6	45.3	41.4	19.0	4.9
8B	PaLM	6.4	24.3	24.1	30.8	30.1	25.0	3.4
	Flan-PaLM	21.9 (+15.5)	49.3	41.3	36.4	31.1	47.5	8.2
62B	PaLM	28.4	55.1	49.0	37.4	43.0	40.5	18.2
	Flan-PaLM	38.8 (+10.4)	59.6	56.9	47.5	44.9	58.7	28.5
540B	PaLM	49.1	71.3	62.9	49.1	63.7	52.9	45.9
	Flan-PaLM	58.4 (+9.3)	73.5	70.9	57.9	66.3	67.8	57.0
62B	cont-PaLM	38.1	61.2	57.6	41.7	53.1	45.7	32.0
	Flan-cont-PaLM	46.7 (+8.6)	66.1	62.0	51.0	53.3	62.7	40.3
540B	U-PaLM	50.2	71.5	64.0	49.2	62.4	54.6	49.9
	Flan-U-PaLM	59.1 (+8.9)	74.1	69.8	59.3	64.9	68.3	60.4

Results of the generality of instruction finetuning on several models

Usability evaluation of open-ended generation

Evaluation of human preferences among open-form responses

- 190 evaluation examples are created: questions about challenging categories of creativity, reasoning over contexts, complex reasoning, planning, and explanation.



- In the zero-shot setting, Flan-PaLM shows preference by a large margin. When using a CoT trigger phrase, the rater preference for Flan-PaLM over PaLM increased by around 10%

Takeaways

- **Scaling curves for instruction finetuning**

- Scaling up the number of templates (1.6K, 3B model or 62K 172B → 1.8K 570B)

→ Both the scaling size of the model and the number of finetuning tasks are expected to continue improving performance.

- **CoT finetuning is critical for reasoning abilities**

- Showing that CoT finetuning a large model improves performance on held-out tasks

→ Whereas prior works, CoT with Flan improves held-out tasks (unseen tasks)

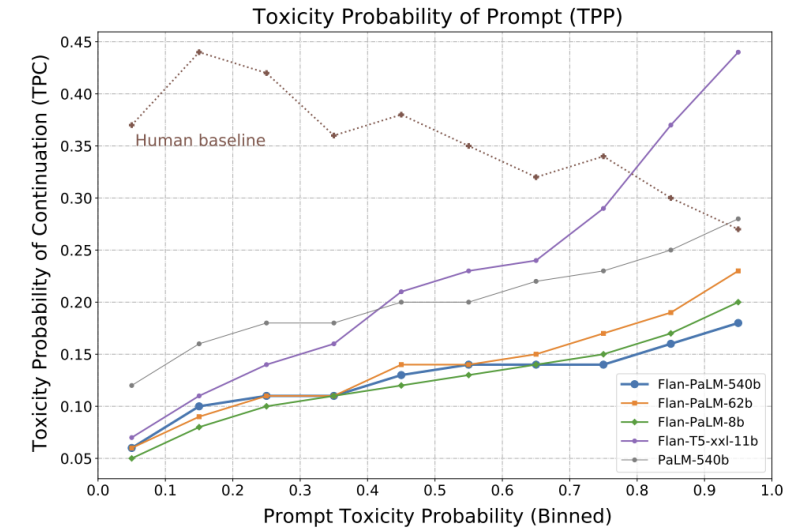
- **Instruction finetuning generalizes across models**

- In section 5, instruction finetuning shows all improvements on different architectures.

→ We could use Flan-T5, Flan-U-PaLM!

Takeaways

- **Instruction finetuning improves usability and mitigates some potential harms**
 - Showing human preference for open-ended evaluations.
- Similarly to Instruction-GPT, finetuned models produce that is better aligned with human preferences.



- **Instruction finetuning is relatively compute-efficient**
 - PaLM 540B instruction finetuning requires only 0.2% of pre-graining compute.
- The Need to develop techniques that might leverage existing checkpoints
- It doesn't change the inference cost of models

Discussion



Haechang : Chain-of-thought Instruction annotation 과정이 자세히 나와있지 않다.



Sanhee : chain-of-thought + instruction 실험을 했을 때 reasoning ability 가 높아지는 결과가 있었는데 chain of thought prompt 퀄리티, 최적 수 엔지니어링 접근에 대해서는 안내와 실험이 부족한 것 같다.

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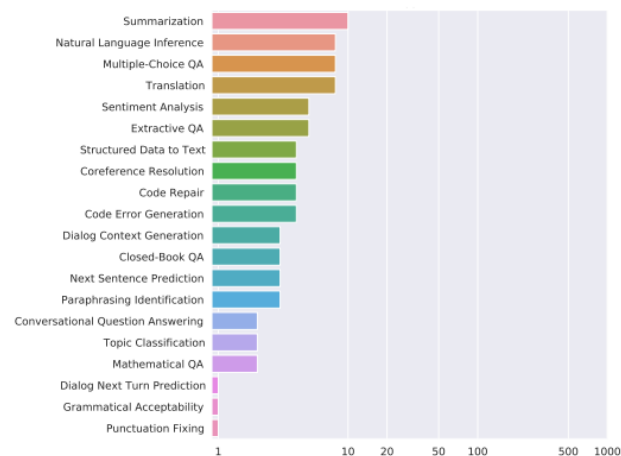
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Q & A

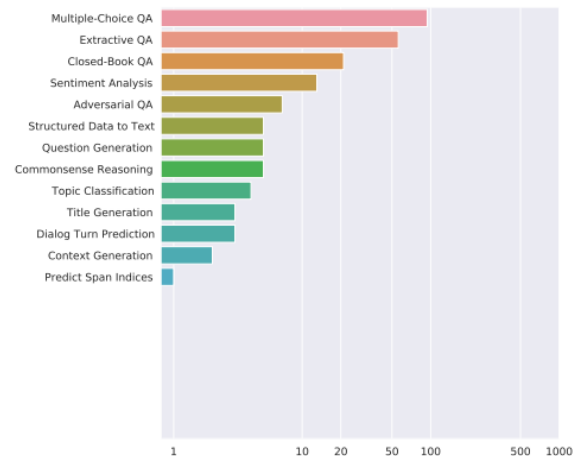
Thank You

Appendix

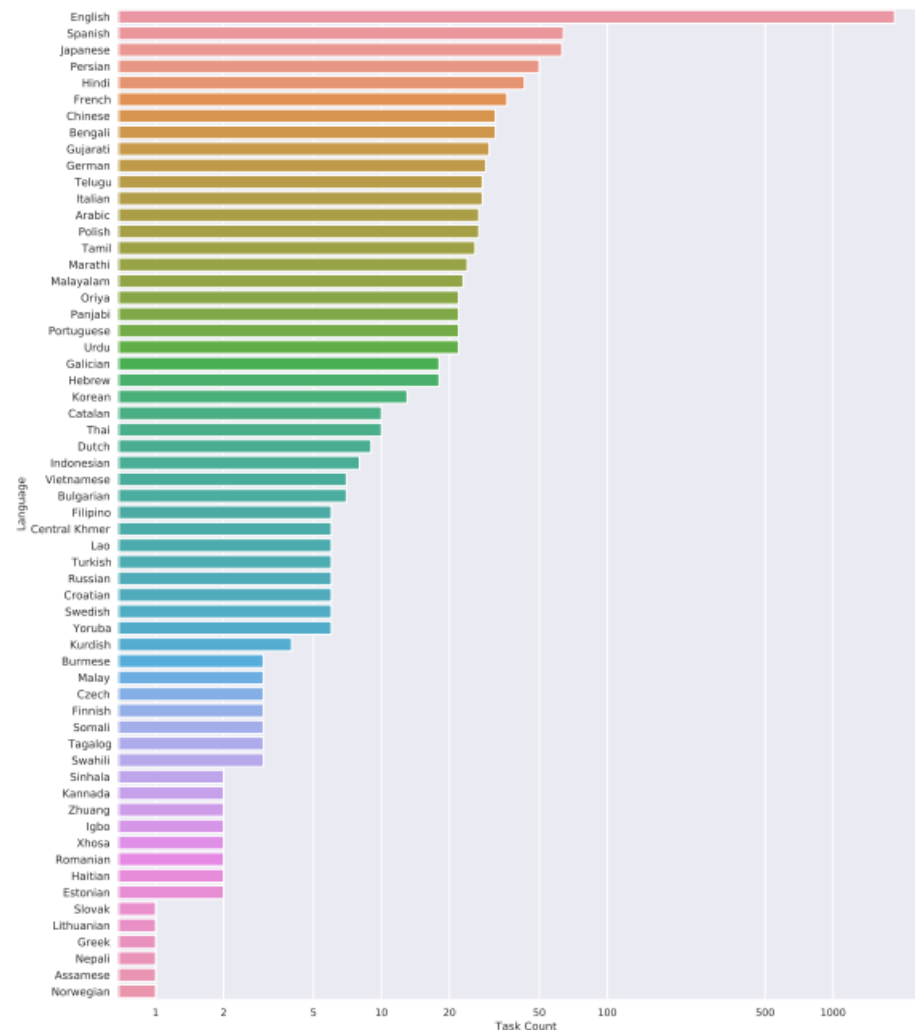
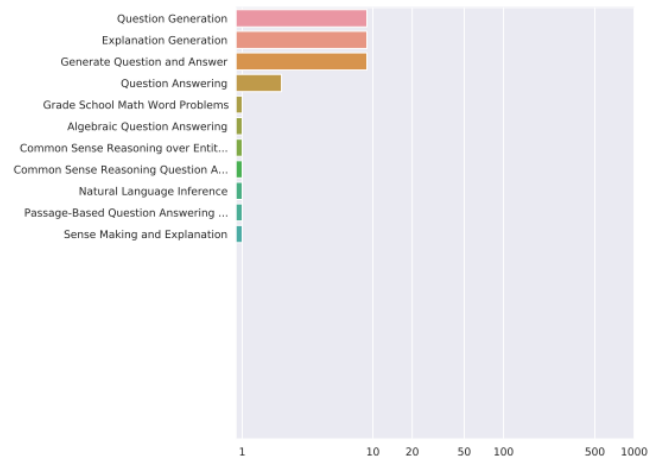
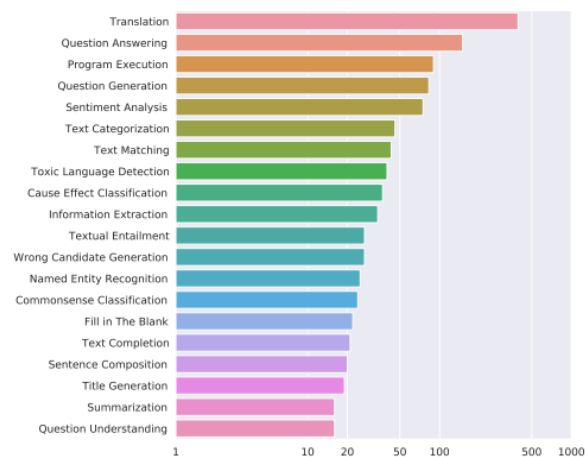
Task ratio per mixtures



(a) Muffin



(b) T0-SF



Appendix

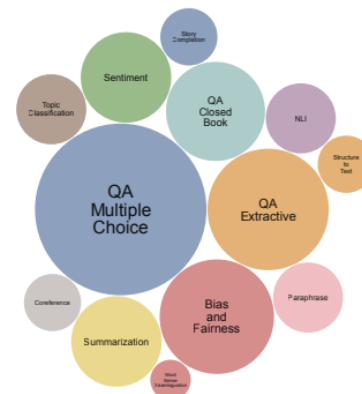
Tasks Mixtures: NIV2



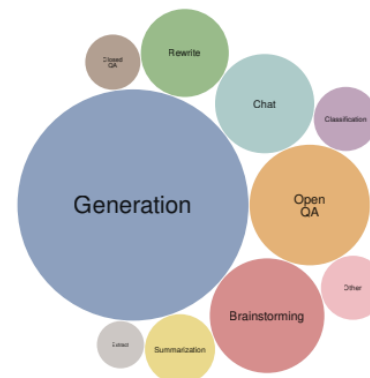
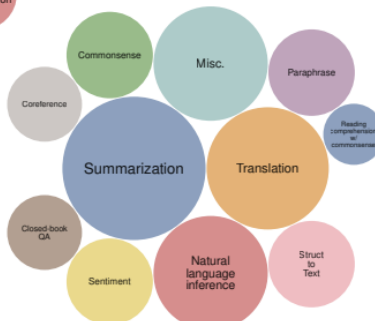
Natural Instruction V2



(b) NATINST



(c) PROMPTSOURCE (T0 subset)



Appendix

I Open-Ended Evaluation Details

The annotation instructions for the human evaluation are provided here:

We have collected responses from different large language models to questions requiring various forms of reasoning. We would like you to help us rank these responses. Each prompt you see will come with responses from (anonymous) large language models, which have been shuffled on EACH ROW, so you the annotator cannot know which model they come from.

PLEASE READ THESE INSTRUCTIONS IN FULL.

Annotation Rules:

- * Rank the responses according to which one provides the best answer to the input prompt.
- * What is the best answer? Make a decision based on (a) the correctness of the answer, and (b) the informativeness of the response. For (a) you are allowed to search the web. Overall, use your best judgment to rank answers based on being the most useful response, which we define as one which is at least somewhat correct, and minimally informative about what the prompt is asking for.
- * If two responses provide the same correctness and informativeness by your judgment, and there is no clear winner, you may rank them the same, but please only use this sparingly.
- * If the answer for a given response is nonsensical, irrelevant, highly ungrammatical/confusing, or does not clearly respond to the given prompt, label it with ‘F’ (for fail) rather than its rank.
- * Long answers are not always the best. Answers which provide succinct, coherent responses may be better than longer ones, if they are at least as correct and informative.

Prompt Style

How do the FLAN instructions differ from GPT-3 or T5 prompts?

GPT-3 prompting is done in a way such that the prompt looks like data that the model has been pretrained on, and the model finishes the continuation. T5 prompts are mostly just a tag for the dataset, which would not work in the zero-shot setting. In contrast, the prompts that we use for FLAN are similar to what would be used to ask a human to perform the task.

For instance, given an input for an NLI task, these would be the prompts.

T5 prompt:

```
cb hypothesis:  At my age you will probably have learnt one lesson.  
premise:  It's not certain how many lessons you'll learn by your  
thirties.
```

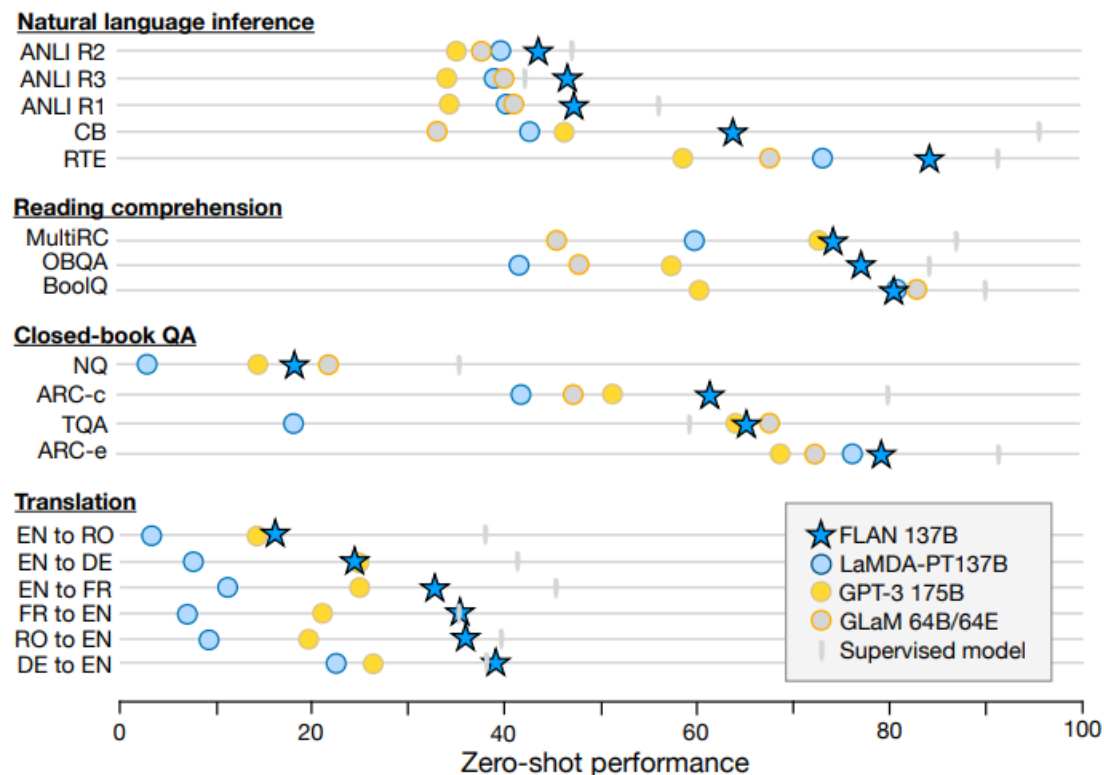
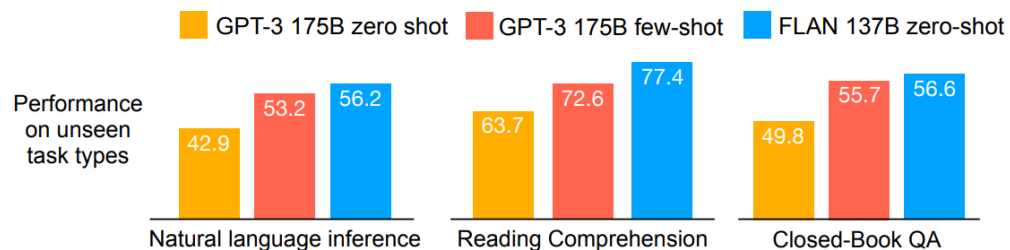
GPT-3 prompt:

```
At my age you will probably have learnt one lesson.  
question:  It's not certain how many lessons you'll learn by your  
thirties.  true, false, or neither?  answer:
```

FLAN prompt:

```
Premise:  At my age you will probably have learnt one lesson.  
Hypothesis:  It's not certain how many lessons you'll learn by your  
thirties.  
Does the premise entail the hypothesis?
```

FLAN (Jason Wei, ICLR 2022) Appendix



Chain-of-Thought (CoT) (Jason Wei, Neurips 2022)

CoT prompting is a gradient-free technique of inducing LLMs to produce intermediate reasoning steps to load the final answer.

PROMPT FOR CSQA

Q: What do people use to absorb extra ink from a fountain pen? Answer Choices: (a) shirt pocket (b) calligrapher's hand (c) inkwell (d) desk drawer (e) blotter

A: The answer must be an item that can absorb ink. Of the above choices, only blotters are used to absorb ink. So the answer is (e).

Q: What home entertainment equipment requires cable?

Answer Choices: (a) radio shack (b) substation (c) television (d) cabinet

A: The answer must require cable. Of the above choices, only television requires cable. So the answer is (c).

Q: The fox walked from the city into the forest, what was it looking for? Answer Choices: (a) pretty flowers (b) hen house (c) natural habitat (d) storybook

A: The answer must be something in the forest. Of the above choices, only natural habitat is in the forest. So the answer is (b).

Q: Sammy wanted to go to where the people were. Where might he go? Answer Choices: (a) populated areas (b) race track (c) desert (d) apartment (e) roadblock

A: The answer must be a place with a lot of people. Of the above choices, only populated areas have a lot of people. So the answer is (a).

Q: Where do you put your grapes just before checking out? Answer Choices: (a) mouth (b) grocery cart (c)super market (d) fruit basket (e) fruit market

A: The answer should be the place where grocery items are placed before checking out. Of the above choices, grocery cart makes the most sense for holding grocery items. So the answer is (b).

Q: Google Maps and other highway and street GPS services have replaced what? Answer Choices: (a) united states (b) mexico (c) countryside (d) atlas

A: The answer must be something that used to do what Google Maps and GPS services do, which is to give directions. Of the above choices, only atlases are used to give directions. So the answer is (d).

Q: Before getting a divorce, what did the wife feel who was doing all the work? Answer Choices: (a) harder (b) anguish (c) bitterness (d) tears (e) sadness

A: The answer should be the feeling of someone getting divorced who was doing all the work. Of the above choices, the closest feeling is bitterness. So the answer is (c).

An exemplar

Few shots

